

\* "Numbers" which we're used to are called Scalars.

• Set of real numbers  $\mathbb{R}$  is a set of scalars.

Vector properties:

• Often use  $\lambda \in \mathbb{R}$  to denote a scalar

Let  $\vec{A}, \vec{B}, \vec{C}$  be vectors,  $\lambda$  a scalar

(i)

$$\lambda(\vec{A} + \vec{B}) = \lambda\vec{A} + \lambda\vec{B} \quad (\text{Distributive})$$

(ii)

$$(\vec{A} \cdot \vec{B}) \cdot \vec{C} = \vec{A} \cdot (\vec{B} \cdot \vec{C}) \quad (\text{Associative})$$

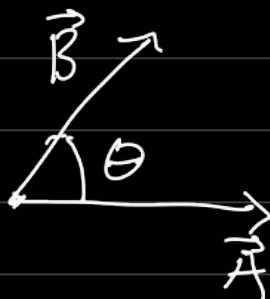
(iii)

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2 = A_x^2 + A_y^2 + A_z^2$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos(\theta)$$

$$(iv) \lambda(\vec{A} \cdot \vec{B}) = (\lambda\vec{A}) \cdot \vec{B} = \vec{A} \cdot (\lambda\vec{B})$$



• There is also a zero vector:

$$\vec{0} = (0, 0, 0)$$

$$(v) \vec{A} + (-\vec{A}) = \vec{0}$$

\* Using our new notation.

We found we could expand a vector in terms of basis vectors

$$\vec{r} = (r_x, r_y, r_z) = r_x \hat{e}_x + r_y \hat{e}_y + r_z \hat{e}_z$$

Since

$$\hat{e}_x = (1, 0, 0) \quad \hat{e}_y = (0, 1, 0) \quad \hat{e}_z = (0, 0, 1)$$

Using  $\vec{A} \cdot \vec{B} = A_x B_x + \dots$  we find

$$\hat{e}_x \cdot \hat{e}_x = 1 = \hat{e}_y \cdot \hat{e}_y = \hat{e}_z \cdot \hat{e}_z$$

$$\hat{e}_x \cdot \hat{e}_y = (1, 0, 0) \cdot (0, 1, 0) = 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 = 0$$

$$\hat{e}_x \cdot \hat{e}_z = 0 \quad (\text{same as above})$$

and likewise for other combinations

• Note:

$$\hat{e}_i \cdot \hat{e}_i = |\hat{e}_i|^2 = 1 \Rightarrow |\hat{e}_i| = 1$$

thus our basis vectors have unit magnitude.

• Note for:  $\vec{a} = a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z$

$$\begin{aligned} \vec{a} \cdot \hat{e}_x &= (a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z) \cdot \hat{e}_x \\ &= a_x \underbrace{\hat{e}_x \cdot \hat{e}_x}_{=1} + a_y \underbrace{\hat{e}_y \cdot \hat{e}_x}_{=0} + a_z \underbrace{\hat{e}_z \cdot \hat{e}_x}_{=0} \end{aligned}$$

therefore

$$\boxed{\vec{a} \cdot \hat{e}_x = a_x}$$

So, the scalar product of a vector with a basis vector  $\hat{e}_i$  gives us the component of that vector along that basis.

Exercise 5.1  $\vec{a} = a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z$   
 $b_x = (\vec{b} \cdot \hat{e}_x), b_y = (\vec{b} \cdot \hat{e}_y), b_z = (\vec{b} \cdot \hat{e}_z)$

then

$$a_x b_x = a_x (\vec{b} \cdot \hat{e}_x) = \vec{b} \cdot (a_x \hat{e}_x)$$

(from property (iv) above)

likewise

$$a_y b_y = a_y (\vec{b} \cdot \hat{e}_y) = \vec{b} \cdot (a_y \hat{e}_y)$$

$$a_z b_z = a_z (\vec{b} \cdot \hat{e}_z) = \vec{b} \cdot (a_z \hat{e}_z)$$

Plugging this into

$$= \vec{b} \cdot (a_x \hat{e}_x) + \vec{b} \cdot (a_y \hat{e}_y) + \vec{b} \cdot (a_z \hat{e}_z)$$

Using property 1(c) or by factoring out  $\vec{b}$ :

$$= \vec{b} \cdot (a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z)$$

$$= \vec{b} \cdot \vec{a}$$

Thus

$$= \vec{b} \cdot \vec{a}$$

Exercise 5.2 Sketch  $\vec{r} \cdot \vec{a} = \text{constant}$ .

• Let us call the constant  $D$ , then

$$\vec{r} \cdot \vec{a} = D$$

$$\vec{r} = (x, y, z) = x \hat{e}_x + y \hat{e}_y + z \hat{e}_z$$

$$\vec{a} = (a_x, a_y, a_z) = a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z$$

→ Important note on  $\vec{r}$ :

$\vec{r}$  is often reserved to be of the form

$$\vec{r} = (x, y, z)$$

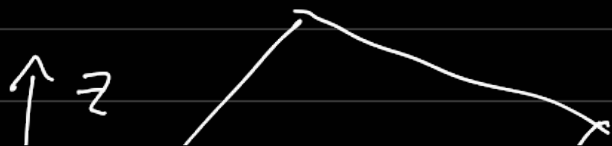
it keeps its components to be the variables  $x, y, z$ .  
It is reserved for measuring the position of some object.

Now,

$$\vec{r} \cdot \vec{a} = x a_x + y a_y + z a_z = D$$

$$x a_x + y a_y + z a_z = D$$

This is the equation of a 2D-plane:





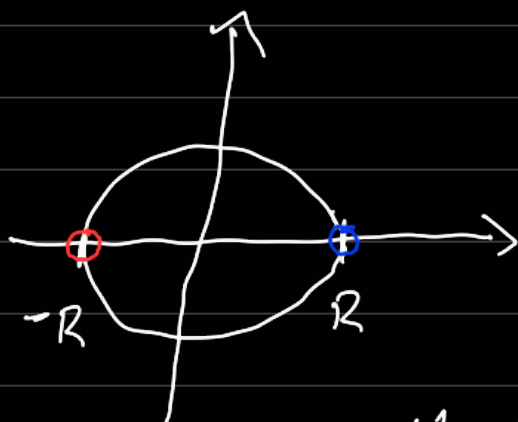
\* To sketch this we can look at different coordinate intercepts:

ex: as a starting point recall the equation of a circle:

$$x^2 + y^2 = R^2$$

when  $y=0$ , this is where the circle crosses the x-axis

$$y=0: x^2 + 0^2 = R^2 \Rightarrow \begin{cases} x = -R \\ x = R \end{cases}$$



likewise we could look at  $x=0$

$$x=0$$

$$y=R$$

$$y=-R$$

This is a trivial example but lets see how it applies.

$$a_x x + a_y y + a_z z = D$$

(i) let  $y=z=0$ , then

$$a_x x = D \Rightarrow x = \frac{D}{a_x}$$

this tells us the plane crosses the  $x$ -axis at  $x = D/a_x$ .

(ii)  $x = y = 0$ , then

$$z = \frac{D}{a_z}$$

(iii)  $x = z = 0$ , then

$$y = D/a_y$$

• So we've gained some geometric information. Next we look at cross-sections

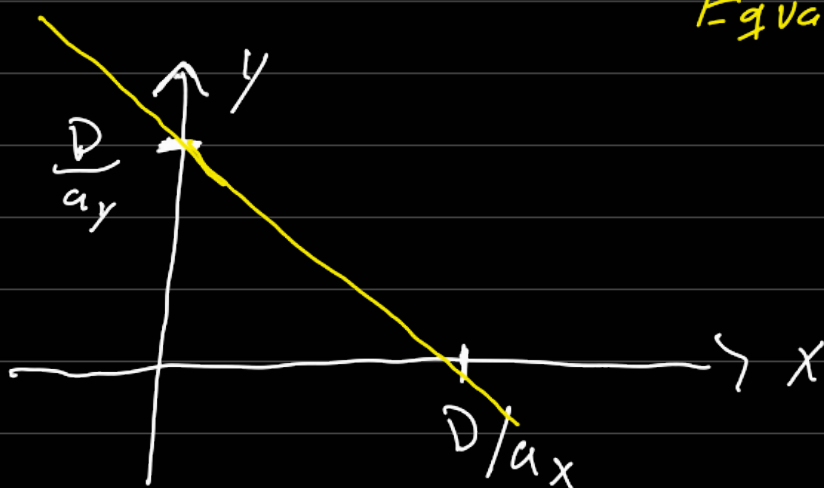
① Let  $z = 0$ : this tells us how the plane crosses the  $x$ - $y$  plane

$$z = 0: a_x x + a_y y = D$$

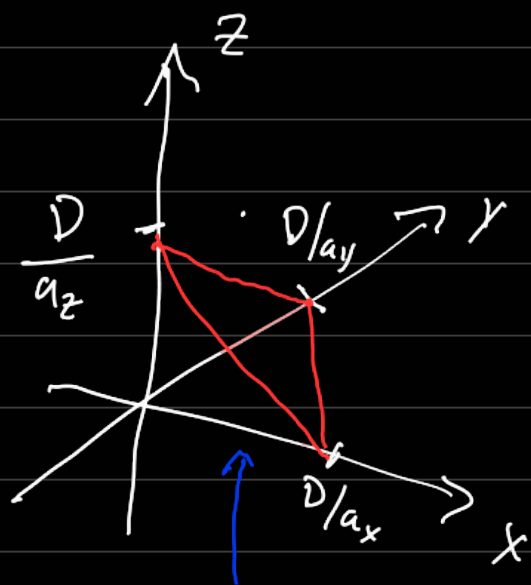
Solve for  $y$ :

$$y = -\frac{a_x}{a_y}x + \frac{D}{a_y}$$

Equation of a line



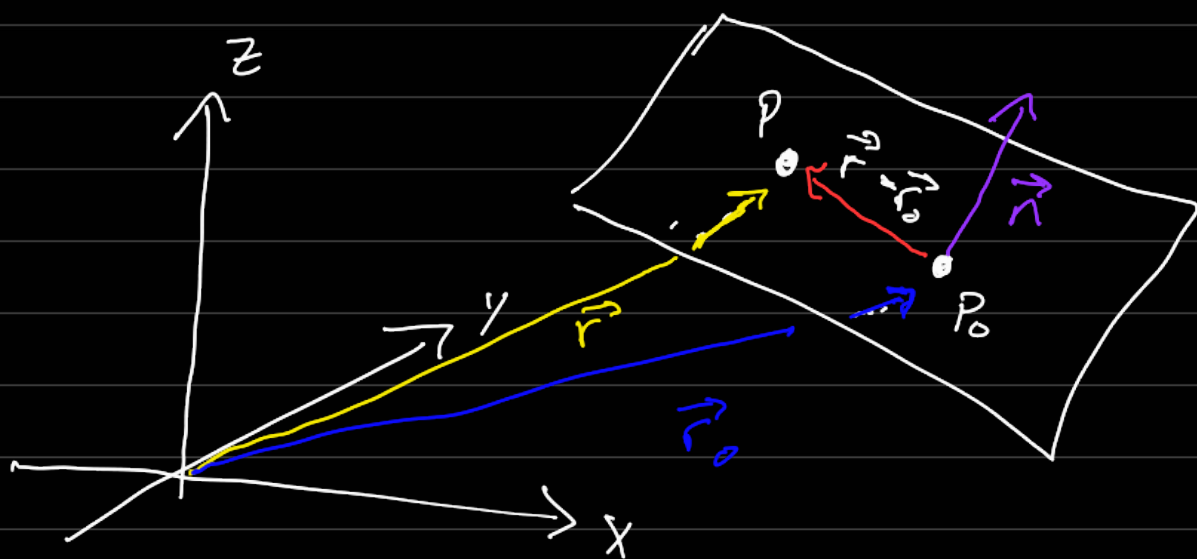
- We can do the same with the other cross-sections.
- Here I draw a small section of the entire plane:



(let me know if this doesn't make sense to you)



\* We used basic ideas to look at the above, let's see how to relate everything:



- Look at two points on the plane  $P_0$  and  $P$ . To each of these points we have vectors

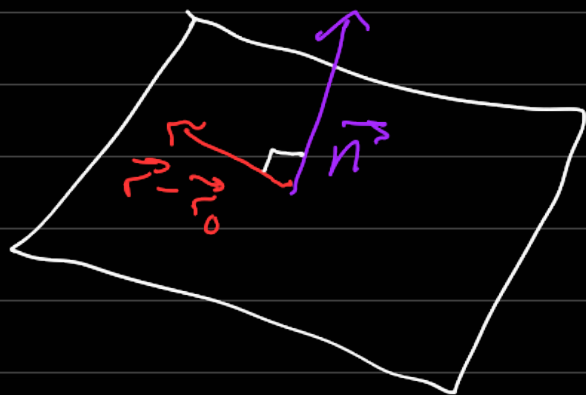
$$P: \vec{r} = (x, y, z)$$

$$P_0: \vec{r}_0 = (x_0, y_0, z_0)$$

- Note that  $(\vec{r} - \vec{r}_0)$  the vector lies on the plane. That is, it is parallel to the plane.

- Now, let  $\vec{n}$  be some vector  $\vec{n} = (a, b, c)$  which

is perpendicular to the line



$$\theta = \frac{\pi}{2}$$

Recall,  $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos(\theta)$

• If we take

$$(\vec{r} - \vec{r}_0) \cdot \vec{n} = |\vec{r} - \vec{r}_0| |\vec{n}| \cos\left(\frac{\pi}{2}\right) = 0$$

$$\begin{aligned} \vec{r} - \vec{r}_0 &= (x, y, z) - (x_0, y_0, z_0) \\ &= (x - x_0, y - y_0, z - z_0) \end{aligned}$$

Thus,

$$(\vec{r} - \vec{r}_0) \cdot \vec{n} = (x - x_0)a + (y - y_0)b + (z - z_0)c = 0$$

Equation of plane 1:

$$(x - x_0)a + (y - y_0)b + (z - z_0)c = 0$$

if we expand

$$ax + by + cz - (ax_0 + by_0 + cz_0) = 0$$

let

$$D \equiv ax_0 + by_0 + cz_0$$

we get

Equation of plane 2:

$$ax + by + cz = 1$$

This is the equation we used above.

\* Notes on the cross product & the (3x3)-determinant:

Recall the cross product is

$$(\vec{a} \times \vec{b}) = (a_y b_z - a_z b_y) \hat{e}_x + (a_z b_x - a_x b_z) \hat{e}_y + (a_x b_y - a_y b_x) \hat{e}_z$$

This can be written as the determinant:

$$\begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

we use the following to "remember" this

$$\begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \underbrace{\begin{vmatrix} a_y & a_z \\ b_y & b_z \end{vmatrix}}_{\text{term 1}} \hat{e}_x - \underbrace{\begin{vmatrix} a_x & a_z \\ b_x & b_z \end{vmatrix}}_{\text{term 2}} \hat{e}_y + \underbrace{\begin{vmatrix} a_x & a_y \\ b_x & b_y \end{vmatrix}}_{\text{term 3}} \hat{e}_z$$

term 1:

$$\begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{vmatrix} a_y & a_z \\ b_y & b_z \end{vmatrix} \hat{e}_x$$

term 2:

$$\begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = - \begin{vmatrix} a_x & a_z \\ b_x & b_z \end{vmatrix} \hat{e}_y$$

$$\begin{vmatrix} \cancel{a_x} \hat{e}_y \hat{e}_z \\ a_x \cancel{a_y} a_z \\ b_x b_y b_z \end{vmatrix} = - \begin{vmatrix} a_x a_z \\ b_x b_z \end{vmatrix} \hat{e}_y$$

Note the = sign!  
this is the only term  
with a minus in front

term 3:

$$\begin{vmatrix} \hat{e}_x \hat{e}_y \hat{e}_z \\ a_x a_y a_z \\ b_x b_y b_z \end{vmatrix} = \begin{vmatrix} a_x a_y \\ b_x b_y \end{vmatrix} \hat{e}_z$$

\* Now  $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$  is the notation for the determinant of a 2x2 matrix.  
it has a simple expansion

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

visually this is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{array}{c} \text{red arrow } \searrow \\ \text{blue arrow } \swarrow \end{array} = ad - bc$$

\* using this expansion we get

term 1:

$$\begin{vmatrix} a_y a_z \\ b_y b_z \end{vmatrix} \hat{e}_x = (a_y b_z - a_z b_y) \hat{e}_x$$

and likewise for the other terms.

## Dyadic product & Matrices:

\* Later on in your maths degree you'll learn this is called a tensor product. To define this using vector (coordinate) notation we must review Matrices:

### \* Matrix:

• An array of numbers which can act on vectors.

### \* Matrix notation:

Recall we found a vector can be written as:

$$\vec{a} = [a_x, a_y, a_z] = a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z$$

now, we introduce one more way of writing a vector

$$\vec{a} \longrightarrow \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix}$$

no commas!  
✓

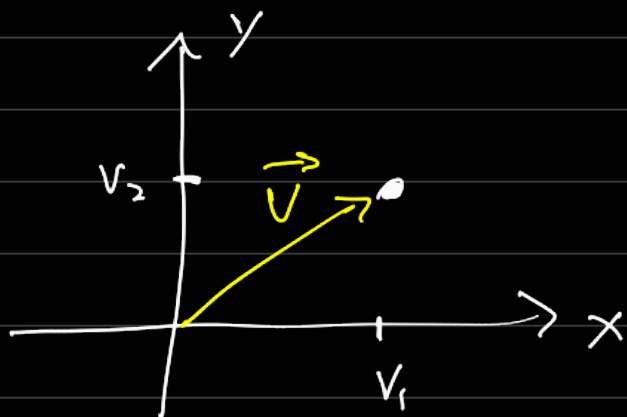
it is written as a column. The reason for this will be apparent shortly:

### \* Matrix acting on a vector:

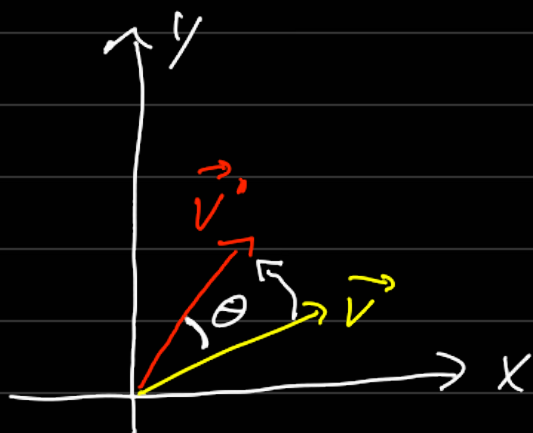
• lets focus on 2D vectors. Let

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

this is some vector in  $\mathbb{R}^2$



- Now, suppose we want to rotate this vector by some angle  $\theta$ :



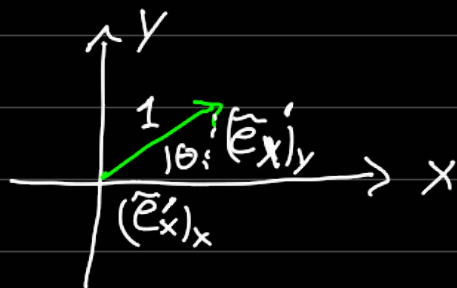
the new vector is different. Why?  
Recall a vector has magnitude & direction  
the magnitude remains unchanged as we  
rotate however the direction has changed

- So, we want to start with a vector  $\vec{v}$  and we want to know what the new vector  $\vec{v}'$  is when we rotate by an angle  $\theta$ .
- Instead of looking at the trig of this system we can instead look at how the basis vectors are transformed

- $\hat{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

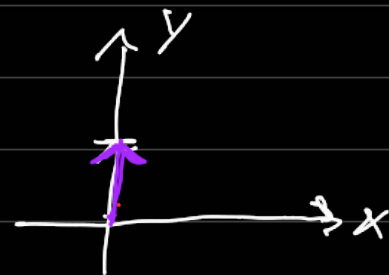


rotate this by  $\theta$ :

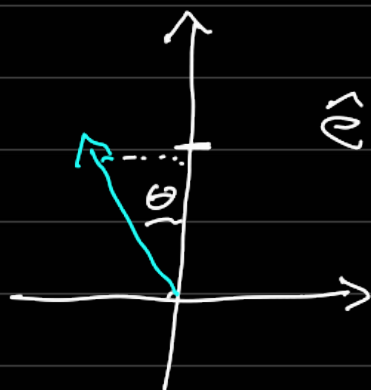


$$\hat{e}_x \rightarrow \tilde{e}_x' = \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix}$$

- likewise for  $\hat{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



rotate by an angle  $\theta$



$$\hat{e}_y \rightarrow \tilde{e}_y' = \begin{pmatrix} -\sin(\theta) \\ \cos(\theta) \end{pmatrix}$$

Note:  $\vec{v} = v_x \hat{e}_x + v_y \hat{e}_y = v_x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + v_y \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$= \begin{pmatrix} v_x \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ v_y \end{pmatrix}$$

$$= \begin{pmatrix} v_x \\ v_y \end{pmatrix}$$

therefore, when we rotate  $v_x$  &  $v_y$  remain the same

$$\vec{v}' = v_x \tilde{e}_x' + v_y \tilde{e}_y'$$

$$= v_x \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} + v_y \begin{pmatrix} -\sin(\theta) \\ \cos(\theta) \end{pmatrix}$$

$$= \begin{pmatrix} v_x \cos(\theta) \\ v_x \sin(\theta) \end{pmatrix} + \begin{pmatrix} -v_y \sin(\theta) \\ v_y \cos(\theta) \end{pmatrix}$$

$$= \begin{pmatrix} V_x \cos(\theta) - V_y \sin(\theta) \\ V_x \sin(\theta) + V_y \cos(\theta) \end{pmatrix}$$

\* Thus, we have a relationship between the components of  $V$  and  $V'$ :

$$\vec{V}' = \begin{pmatrix} V_x' \\ V_y' \end{pmatrix} = \begin{pmatrix} V_x \cos(\theta) - V_y \sin(\theta) \\ V_x \sin(\theta) + V_y \cos(\theta) \end{pmatrix}$$

or

$$V_x' = V_x \cos(\theta) + V_y \sin(\theta)$$

$$V_y' = V_x \sin(\theta) + V_y \cos(\theta)$$

\* Okay, so this may seem like a long aside but lets tie everything together:

The above answer looks quite suggestive. We can write it in matrix notation

$$\vec{V}' = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} V_x \\ V_y \end{pmatrix} = \begin{pmatrix} V_x' \\ V_y' \end{pmatrix}$$

We call

$$R = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

the rotation matrix as it takes a vector  $V$  and rotates it to a vector  $V'$ .

• In matrix notation:

$$\vec{V}' = R \vec{V}$$

• Although I have introduced matrices as a

rotation of a vector the basic idea is this:

Matrices act as vector transformations

a matrix takes a vector and transforms it into another

- Another example of this is the identity which does "nothing"

$$\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Since

$$\begin{aligned} \mathbb{I} \vec{v} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 1v_x + 0v_y \\ 0v_x + 1v_y \end{pmatrix} \\ &= \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \vec{v} \quad \checkmark \end{aligned}$$

\* We can define the transpose of a matrix  
 $M^T$  ← write a T for this

but we also define the transpose of a vector  $\vec{v}$

$$v^T = \begin{pmatrix} v_x \\ v_y \end{pmatrix}^T \equiv (v_x \ v_y)$$

this is now a row vector. Note that this is NOT the same as our notation from earlier  $\vec{v} = (v_x, v_y)$

Note the comma → whereas the transpose above has no commas

\* Important: I have assumed you have familiarity with the basics of Matrices. If you are not then let me know

\* Using our new notation we can rewrite the scalar product:

$$\vec{a} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix}$$

$$\vec{a}^T = (a_x \ a_y \ a_z), \quad \vec{b}^T = (b_x \ b_y \ b_z)$$

then

$$\vec{a}^T \vec{b} = (a_x \ a_y \ a_z) \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix}$$

$$= a_x b_x + a_y b_y + a_z b_z$$

but this is equal to  $\vec{a} \cdot \vec{b}$ , so

$$\boxed{\vec{a} \cdot \vec{b} = \vec{a}^T \vec{b}}$$

(prove to yourself that  $\vec{a}^T \vec{b} = \vec{b}^T \vec{a}$ )

\* Now ready to define the Dyadic (tensor) product of vectors  $\vec{a}$  and  $\vec{b}$  as

$$\boxed{\vec{a} \vec{b}^T = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} (b_x \ b_y \ b_z) = \vec{a} \otimes \vec{b}}$$

(you'll see this written as  $\vec{a} \otimes \vec{b}$  later in uni.)  
this is computed as:

$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} (b_x \ b_y \ b_z) = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \end{pmatrix}$$

$$\begin{matrix} \overbrace{a_z}^{(a_z)} \\ \underbrace{3 \times 1} \quad \underbrace{1 \times 3} \end{matrix}$$

$$\begin{matrix} a_z b_x & a_z b_y & a_z b_z \\ \underbrace{\hspace{10em}} \end{matrix}$$

3x3 - Matrix

- Lets look at the first examples in your notes

$$\hat{e}_x \circ \hat{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} (1 \ 0 \ 0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\hat{e}_z \circ \hat{e}_y = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} (0 \ 2 \ 0) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

etc...

- Do the other examples in the notes.

\* Lets derive the relationship

$$(\vec{b} \circ \vec{c})(\vec{r}) = \vec{b}(\vec{c} \cdot \vec{r})$$

proof:  $\vec{b} \circ \vec{c} = \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix} (c_x \ c_y \ c_z) = \begin{pmatrix} b_x c_x & b_x c_y & b_x c_z \\ b_y c_x & b_y c_y & b_y c_z \\ b_z c_x & b_z c_y & b_z c_z \end{pmatrix}$

Now,

$$(\vec{b} \circ \vec{c}) \vec{r} = \begin{pmatrix} b_x c_x & b_x c_y & b_x c_z \\ b_y c_x & b_y c_y & b_y c_z \\ b_z c_x & b_z c_y & b_z c_z \end{pmatrix} \begin{pmatrix} r_x \\ r_y \\ r_z \end{pmatrix}$$

$$= \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

$$\begin{aligned} \alpha &= b_x c_x r_x + b_x c_y r_y + b_x c_z r_z \\ &= b_x (r_x c_x + r_y c_y + r_z c_z) \\ &= b_x (\vec{r} \cdot \vec{c}) \end{aligned}$$

likewise for  $\beta, \gamma$  we find

$$\beta = b_y (\vec{r} \cdot \vec{c})$$

$$\gamma = b_z (\vec{r} \cdot \vec{c})$$

so

$$(\vec{b} \cdot \vec{c}) (\vec{r}) = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} b_x (\vec{r} \cdot \vec{c}) \\ b_y (\vec{r} \cdot \vec{c}) \\ b_z (\vec{r} \cdot \vec{c}) \end{pmatrix}$$

recall  $\vec{r} \cdot \vec{c}$  is a number so we can factor this out

$$(\vec{b} \cdot \vec{c}) \vec{r} = \vec{r} \cdot \vec{c} \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix} = \vec{b} (\vec{c} \cdot \vec{r})$$

$$\underbrace{\begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix}}_{\vec{b}}$$

$$\boxed{(\vec{b} \cdot \vec{c}) \vec{r} = \vec{b} (\vec{c} \cdot \vec{r})}$$

as claimed in your notes.

